

Lab Sheet 2

Database Management System Lab (BTCS203P)

B. TECH AI&ML, CYBER SECURITY, Data Science 3RD Semester (Session 2026-2025)

Objective:

To understand and practice Data Manipulation Language (DML) commands for creating and modifying the structure of a table in a relational database using SQL (MySQL).

Prerequisites:

1. Basic knowledge of SQL.
2. MySQL installed and set up.

Lab Sheet 2: DML Commands – INSERT, UPDATE and DELETE

Perform the following SQL tasks using suitable examples:

1. Insert one record into the Student table.

Insert into Student values('Yash', 22, 'Data Science', '2004-07-26', 'B.Tech', 2022, 'Meerut');

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Sname	Age	Branch	DOB	Course	Year_of_Admission	City
	Yash	22	Data Science	2004-07-26	B.Tech	2022	Meerut

2. Insert five student records using separate INSERT statements.

```

Insert into Student values('Pooja', 19, 'Civil', '2007-08-11', 'B.Tech', 2025);
-- Insert into Student values('Arjun', 22, 'Mechanical', '2004-05-30', 'B.Tech', 2022);
-- Insert into Student values('Kavya', 20, 'AI', '2006-09-21', 'B.Tech', 2024);
-- Insert into Student values('Nikhil', 21, 'ECE', '2005-02-17', 'B.Tech', 2023);
-- Insert into Student values('Mehak', 19, 'CSE', '2007-11-03', 'B.Tech', 2025);
-- Select * from Student;

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Sname	Age	Branch	DOB	Course	Year_of_Admission	City
▶	Yash	22	Data Science	2004-07-26	B.Tech	2022	Meerut
	Pooja	19	Civil	2007-08-11	B.Tech	2025	Meerut
	Arjun	22	Mechanical	2004-05-30	B.Tech	2022	Delhi
	Kavya	20	AI	2006-09-21	B.Tech	2024	Mumbai
	Nikhil	21	ECE	2005-02-17	B.Tech	2023	Delhi

Student 5

×

3. Insert multiple records into the Student table using a single statement.

```

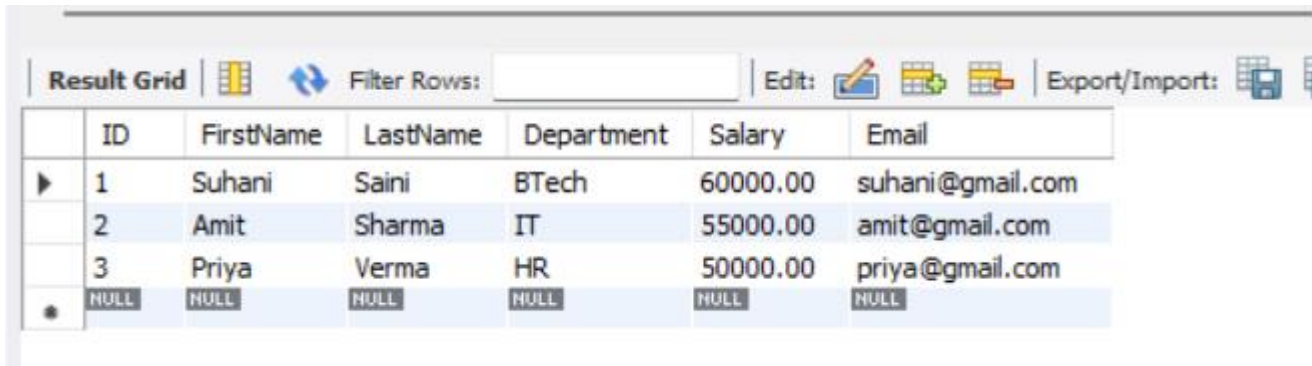
Insert into Student (Sname, Age, Branch, DOB, Course, Year_of_Admission, City)
VALUES
('Pooja', 19, 'Civil', '2007-08-11', 'B.Tech', 2025, 'Meerut'),
('Arjun', 22, 'Mechanical', '2004-05-30', 'B.Tech', 2022, 'Delhi'),
('Kavya', 20, 'AI', '2006-09-21', 'B.Tech', 2024, 'Mumbai'),
('Nikhil', 21, 'ECE', '2005-02-17', 'B.Tech', 2023, 'Delhi'),

```

('Mehak', 19, 'CSE', '2007-11-03', 'B.Tech', 2025, 'Roorkee');

4. Insert records into the Employee table.

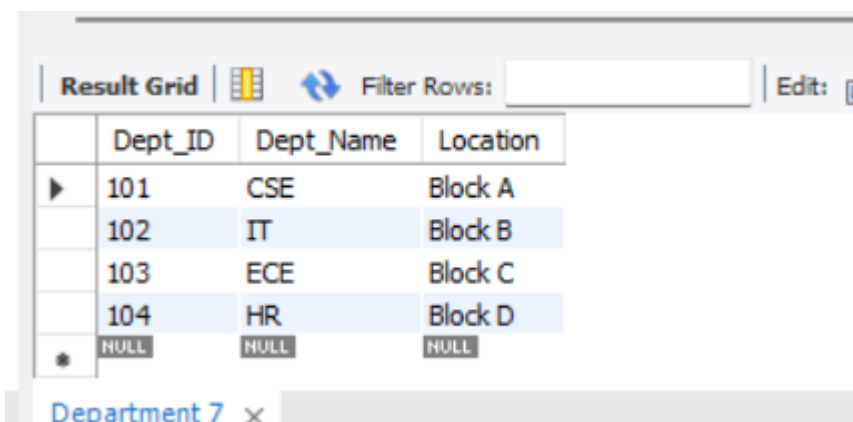
```
INSERT INTO Employee (FirstName, LastName, Department, Salary, Email)
VALUES
('Suhani', 'Saini', 'BTech', 60000, 'suhani@gmail.com'),
('Amit', 'Sharma', 'IT', 55000, 'amit@gmail.com'),
('Priya', 'Verma', 'HR', 50000, 'priya@gmail.com');
Select * from Employee;
```



	ID	FirstName	LastName	Department	Salary	Email
▶	1	Suhani	Saini	BTech	60000.00	suhani@gmail.com
	2	Amit	Sharma	IT	55000.00	amit@gmail.com
	3	Priya	Verma	HR	50000.00	priya@gmail.com
•	NULL	NULL	NULL	NULL	NULL	NULL

5. Insert records into the Department table.

```
CREATE TABLE Department (
    Dept_ID INT AUTO_INCREMENT PRIMARY KEY,
    Dept_Name VARCHAR(50) NOT NULL,
    Location VARCHAR(50)
);
INSERT INTO Department (Dept_ID, Dept_Name, Location)
VALUES
(101, 'CSE', 'Block A'),
(102, 'IT', 'Block B'),
(103, 'ECE', 'Block C'),
(104, 'HR', 'Block D');
Select * From Department;
```



	Dept_ID	Dept_Name	Location
▶	101	CSE	Block A
	102	IT	Block B
	103	ECE	Block C
	104	HR	Block D
•	NULL	NULL	NULL

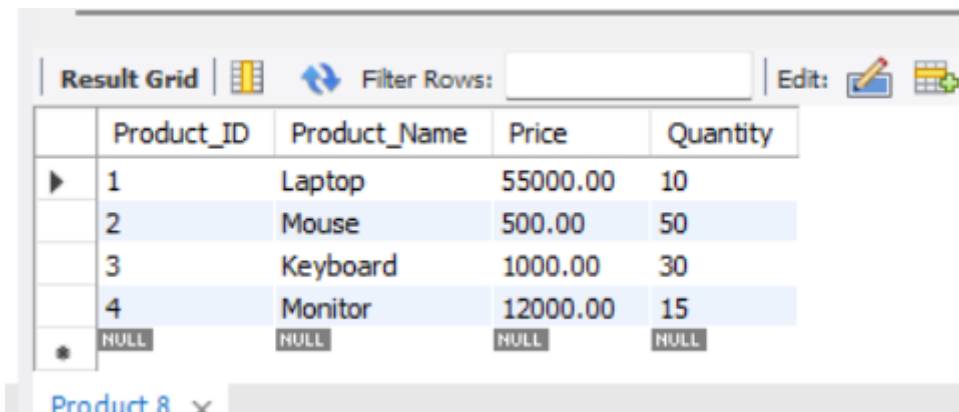
Department 7 x

6. Insert records into the Product table.

```

CREATE TABLE Product (
    Product_ID INT AUTO_INCREMENT PRIMARY KEY,
    Product_Name VARCHAR(100) NOT NULL,
    Price DECIMAL(10,2),
    Quantity INT
);
INSERT INTO Product (Product_ID, Product_Name, Price, Quantity)
VALUES
(1, 'Laptop', 55000, 10),
(2, 'Mouse', 500, 50),
(3, 'Keyboard', 1000, 30),
(4, 'Monitor', 12000, 15);
Select * From Product;

```



The screenshot shows a database interface with a 'Result Grid' tab. It displays the data for the 'Product' table. The table has four columns: Product_ID, Product_Name, Price, and Quantity. There are four data rows and one row for NULL values.

	Product_ID	Product_Name	Price	Quantity
▶	1	Laptop	55000.00	10
	2	Mouse	500.00	50
	3	Keyboard	1000.00	30
	4	Monitor	12000.00	15
•	NULL	NULL	NULL	NULL

Product 8

7. Insert records into the Customer table.

```

INSERT INTO Course (Course_ID, Course_Name, Duration, Fees)
VALUES
(101, 'BTech CSE', '4 Years', 120000),
(102, 'BTech IT', '4 Years', 115000),
(103, 'BTech Data Science', '4 Years', 130000),
(104, 'BCA', '3 Years', 90000);

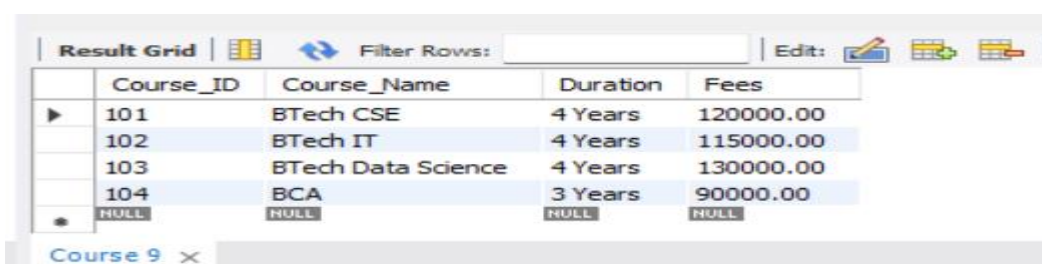
```

8. Insert records into the Course table.

```

INSERT INTO Course (Course_ID, Course_Name, Duration, Fees)
VALUES
(101, 'BTech CSE', '4 Years', 120000),
(102, 'BTech IT', '4 Years', 115000),
(103, 'BTech Data Science', '4 Years', 130000),
(104, 'BCA', '3 Years', 90000);

```



The screenshot shows a database interface with a 'Result Grid' tab. It displays the data for the 'Course' table. The table has four columns: Course_ID, Course_Name, Duration, and Fees. There are four data rows and one row for NULL values.

	Course_ID	Course_Name	Duration	Fees
▶	101	BTech CSE	4 Years	120000.00
	102	BTech IT	4 Years	115000.00
	103	BTech Data Science	4 Years	130000.00
	104	BCA	3 Years	90000.00
•	NULL	NULL	NULL	NULL

Course 9

9. Insert a student record with values for all columns.

```
INSERT INTO Student (Sname, age, branch)
VALUES ('Riya', 19, 'CSE');
Select * from Student;
```

10. Insert a student record by specifying only selected columns.

```
UPDATE Student
SET Year_of_Admission=2022
WHERE Sname = 'Rahul';
```

11. Update the city of a particular student.

12. Update the salary of a particular employee.

```
UPDATE Employee
SET Salary = 65000
WHERE ID = 1;
```

	ID	FirstName	LastName	Department	Salary	Email
▶	1	Suhani	Saini	BTech	65000.00	suhani@gmail.com
	2	Amit	Sharma	IT	55000.00	amit@gmail.com
	3	Priya	Verma	HR	50000.00	priya@gmail.com
•	NULL	NULL	NULL	NULL	NULL	NULL

13. Increase the salary of all employees by 10%.

```
UPDATE Employee
SET Salary = Salary * 1.10;
```

14. Increase the price of all products by 5%.

15. Change the course of a particular student.

```
UPDATE Student
SET Course = 'BTech CSE'
WHERE Sname = 'Rahul';
```

16. Update the department of an employee.

```
UPDATE Employee
SET Department = 'CSE'
WHERE ID = 1;
```

Result Grid			
	Dept_ID	Dept_Name	Location
▶	101	CSE	Block A
	102	IT	Block B
	103	ECE	Block C
	104	HR	Block D
•	NULL	NULL	NULL

17. Update the city of all students from Meerut to Delhi.

```
UPDATE Student
SET City = 'Delhi'
WHERE City = 'Meerut';
```

18. Update the fees of a particular course.

```
UPDATE Course
SET Fees = 140000
WHERE Course_Name = 'BTech Data Science';
```

Course_ID	Course_Name	Duration	Fees
101	BTech CSE	4 Years	120000.00
102	BTech IT	4 Years	115000.00
103	BTech Data Science	4 Years	130000.00
104	BCA	3 Years	90000.00
NULL	NULL	NULL	NULL

19. Update multiple columns of a student simultaneously.

20. Delete a particular student record using Student_ID

```
DELETE FROM Student
WHERE Year_of_Admission=2025;
```

21. Delete all students belonging to a particular city.

```
DELETE FROM Student
WHERE City = 'Meerut';
```

22. Delete employees whose salary is below a specified amount.

```
DELETE FROM Employee
WHERE Salary < 50000;
```

23. Delete products whose quantity is zero.

```
DELETE FROM Product
WHERE Quantity = 0;
```

24. Delete records satisfying multiple conditions using AND/OR.

25. Perform a complete sequence of INSERT, UPDATE and DELETE operations on a Student table and display the table after each operation.

```
INSERT INTO Student
VALUES ('Rahul', 21, 'CSE', '2005-04-15', 'B.Tech', 2023, 'Jaipur');
```

```
SELECT * FROM Student;
```

Sname	Age	Branch	DOB	Course	Year_of_Admission	City
Yash	22	Data Science	2004-07-26	B.Tech	2022	Meerut
Pooja	19	Civil	2007-08-11	B.Tech	2022	Meerut
Arjun	22	Mechanical	2004-05-30	B.Tech	2023	Delhi
Kavya	20	AI	2006-09-21	B.Tech	2024	Mumbai
Nikhil	21	ECE	2005-02-17	B.Tech	2023	Delhi

```
UPDATE Student
SET City = 'Delhi'
WHERE Sname = 'Yash';
```

```
SELECT * FROM Student;
```

Result Grid							
		Filter Rows:		Export:		Wrap Cell Content:	
	Sname	Age	Branch	DOB	Course	Year_of_Admission	City
►	Yash	22	Data Science	2004-07-26	B.Tech	2022	Meerut
	Pooja	19	Civil	2007-08-11	B.Tech	B.Tech	Meerut
	Arjun	22	Mechanical	2004-05-30	B.Tech	2022	Delhi
	Kavya	20	AI	2006-09-21	B.Tech	2024	Mumbai
	Nikhil	21	ECE	2005-02-17	B.Tech	2023	Delhi

```
DELETE FROM Student
WHERE Sname = 'Rahul';
```

```
SELECT * FROM Student;
```

